

2(a)	change in velocity / time (taken)	B1
2(b)(i)	air resistance increases (with speed/with time)	B1
	resultant force decreases (as speed increases/with time) so acceleration decreases (as speed increases/with time)	B1
	when air resistance equals the weight the speed/velocity/ v becomes constant	B1
2(b)(ii)	speed = 36 m s^{-1}	A1
2(b)(iii)	height given by area under the curve	C1
	height = 950 m <i>Round to two significant figures and award 2 marks for a value in the range 920–980 m and 1 mark for a value in the range 900–910 m or 990–1000 m.</i>	A2
2(b)(iv)	line starting at (0, 9.8)	B1
	curve with negative gradient between $t = 0$ and $t = 20 \text{ s}$	B1
	line showing zero acceleration between $t = 20 \text{ s}$ and $t = 30 \text{ s}$	B1